

Technical Document #998: Cloud Computing - Comprehensive Overview

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Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.
- Multi-cloud strategies use services from multiple cloud providers.
- Containerization packages applications with their dependencies.